

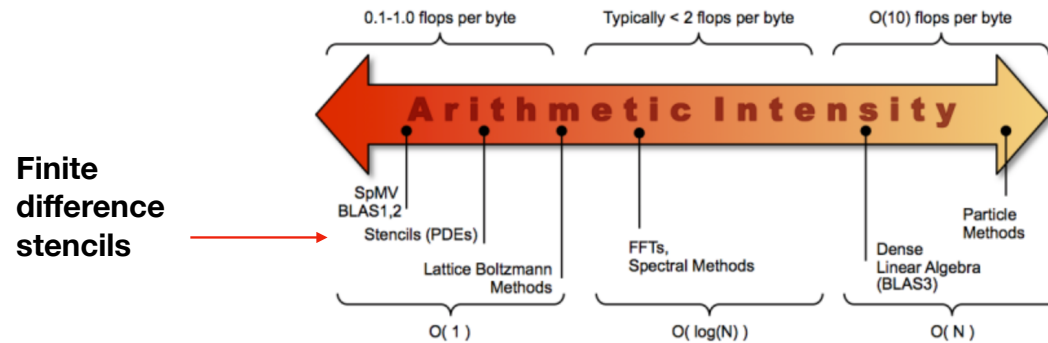
Algorithmic intensity and the roof-line model

Introduction

- To understand performance we need to consider the characteristics of the algorithms we are using
- Different algorithms have a different ratio between the number of floating point operations required and the number of memory accesses required
- We can express this as an “arithmetic intensity” (also called “algorithmic intensity” or “operational intensity”)

Arithmetic intensity

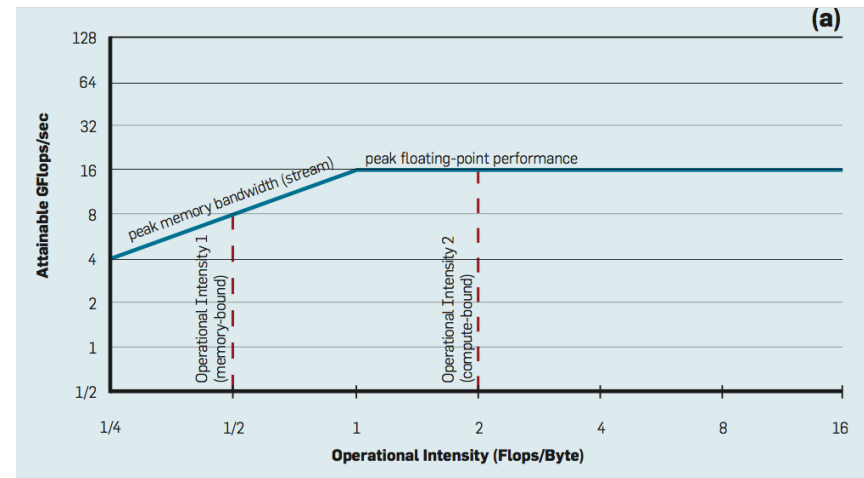
- The arithmetic intensity is the ratio of floating point operations to data movement (units are flops per bytes)
- Different algorithms have different arithmetic intensities
- The arithmetic intensity may also depend on the problem size



<https://crd.lbl.gov/departments/computer-science/PAR/research/roofline/>

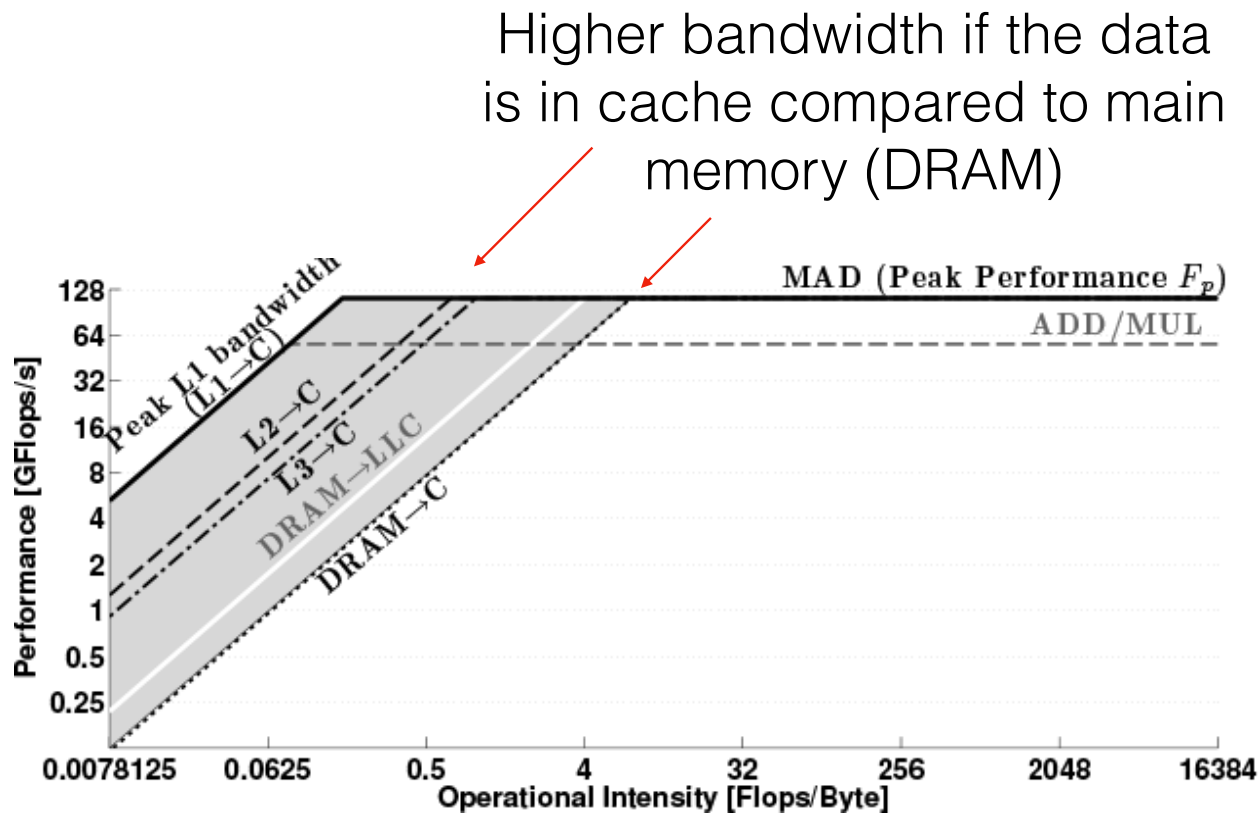
Roofline model

- The roofline model tells us about floating point performance based on peak performance, memory bandwidth and arithmetic intensity
- An application may be **compute bound** or **memory bound** depending on the algorithm and hardware



S. Williams, A. Waterman, D. Patterson, "Roofline: an insightful visual performance model for multicore architectures", Communications of the ACM (CACM), April 2009, [doi:10.1145/1498765.1498785](https://doi.org/10.1145/1498765.1498785)

Roofline model



A. Ilic, F. Pratas and L. Sousa, "Cache-aware Roofline model: Upgrading the loft," in IEEE Computer Architecture Letters, vol. 13, no. 1, pp. 21-24, 21 Jan.-June 2014, [doi:10.1109/L-CA.2013.6](https://doi.org/10.1109/L-CA.2013.6)

Unit Summary

- Different algorithms have different arithmetic intensities (floating point operations per byte of data)
- Lower arithmetic intensity means that the algorithm is limited by memory bandwidth (memory bound)
- Higher arithmetic intensity algorithms can make more efficient use of floating point hardware and are limited by the floating point performance (compute bound)
- We can use the roof-line model to understand how the algorithm performs on a particular hardware platform